Rotational Motion (Supplementary Problems)

1. What is the angular displacement of each of the following hands of a clock in 1 h? State your answer in three significant digits.
a. the second hand
b. the minute hand c. the hour hand

2. If a truck has a linear acceleration of 1.85 m/s^2 and the wheels have an angular acceleration of 5.23 rad/s^2, what is the diameter of the truck's wheels?

3. The truck in the previous problem is towing a trailer with wheels that have a diameter of 48 cm.
a. How does the linear acceleration of the trailer compare with that of the truck?
b. How do the angular accelerations of the wheels of the trailer and the wheels of the truck compare?

4. You want to replace the tires on your car with tires that have a larger diameter. After you change the tires, for trips at the same speed and over the same distance, how will the angular velocity and number of revolutions change?

5. Consider the wrench in Example Problem 1. What force is needed if it is applied to the wrench at a point perpendicular to the wrench?

6. If a torque of 55.0 Nm is required and the largest force that can be exerted by you is 135 N, what is the length of the lever arm that must be used?

7. You have a 0.234-m-long wrench. A job requires a torque of 32.4 Nm, and you can exert a force of 232 N. What is the smallest angle, with respect to the vertical, at which the force can be exerted?

8. You stand on the pedal of a bicycle. If you have a mass of 65 kg, the pedal makes an angle of 35° above the horizontal, and the pedal is 18 cm from the center of the chain ring, how much torque would you exert?

9. If the pedal in problem 8 is horizontal, how much torque would you exert? How much torque would you exert when the pedal is vertical?

10. Ashok, whose mass is 43 kg, sits 1.8 m from the center of a seesaw. Steve, whose mass is 52 kg, wants to balance Ashok. How far from the center of the seesaw should Steve sit?

11. A bicycle-chain wheel has a radius of 7.70 cm. If the chain exerts a 35.0-N force on the wheel in the clockwise direction, what torque is needed to keep the wheel from turning?

12. Two baskets of fruit hang from strings going around pulleys of different diameters, as shown in Figure 8-6. What is the mass of basket A?

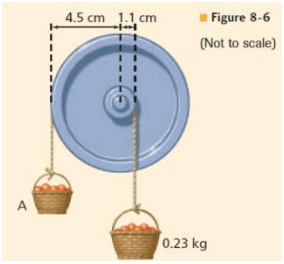


Figure 8-6
(Not to scale)

13. Suppose the radius of the larger pulley in problem 18 was increased to 6.0 cm. What is the mass of basket A now? 20. A bicyclist, of mass 65.0 kg, stands on the pedal of a bicycle. The crank, which is 0.170 m long, makes a 45.0° angle with the vertical, as shown in Figure 8-7. The crank is attached to the chain wheel, which has a radius of 9.70 cm. What force must the chain exert to keep the wheel from turning?

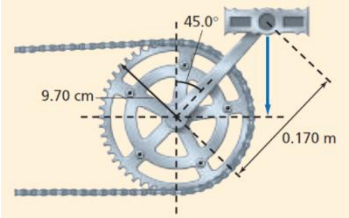


Figure 8-7

14. Two children of equal masses sit 0.3 m from the center of a seesaw. Assuming that their masses are much greater than that of the seesaw, by how much is the moment of inertia increased when they sit 0.6 m from the center?

15. Suppose there are two balls with equal diameters and masses. One is solid, and the other is hollow, with all its mass distributed at its surface. Are the moments of inertia of the balls equal? If not, which is greater?

16. Figure 8-9 shows three massive spheres on a rod of very small mass. Consider the moment of inertia of the system, first when it is rotated about sphere A, and then when it is rotated about sphere C. Are the moments of inertia the same or different? Explain. If the moments of inertia are different, in which case is the moment of inertia greater?

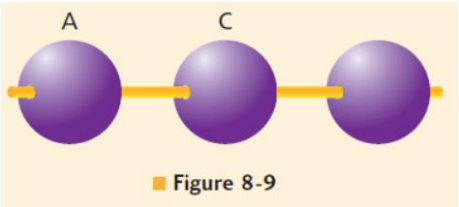


Figure 8-9

17. Each sphere in the previous problem has a mass of 0.10 kg. The distance between spheres A and C is 0.20 m. Find the moment of inertia in the following instances: rotation about sphere A, rotation about sphere C.

18. Torque A solid steel wheel has a mass of 15 kg and a diameter of 0.44 m. It starts at rest. You want to make it rotate at 8.0 rev/s in 15 s.
a. What torque must be applied to the wheel?
b. If you apply the torque by wrapping a strap around the outside of the wheel, how much force should you exert on the strap?

19. Consider the wheel in Example Problem 18. If the force on the strap were twice as great, what would be the speed of rotation of the wheel after 15 s?

20. A solid wheel accelerates at 3.25 rad/s^2 when a force of 4.5 N exerts a torque on it. If the wheel is replaced by a wheel with all of its mass on the rim, the moment of inertia is given by $I = mr^2$. If the same angular velocity were desired, what force would have to be exerted on the strap?

21. A bicycle wheel can be accelerated either by pulling on the chain that is on the gear or by pulling on a string wrapped around the tire. The wheel's radius is 0.38 m, while the radius of the gear is 0.14 m. If you obtained the needed acceleration with a force of 15 N on the chain, what force would you need to exert on the string?

22. The bicycle wheel in problem 21 is used with a smaller gear whose radius is 0.11 m. The wheel can be accelerated either by pulling on the chain that is on the gear or by pulling string that is wrapped around the tire. If you obtained the needed acceleration with a force of 15 N on the chain, what force would you need to exert on the string?

23. A disk with a moment of inertia of 0.26 kgm^2 is attached to a smaller disk mounted on the same axle. The smaller disk has a diameter of 0.180 m and a mass of 2.5 kg. A strap is wrapped around the smaller disk, as shown in Figure 8-10. Find the force needed to give this system an angular acceleration of 2.57 rad/s^2.

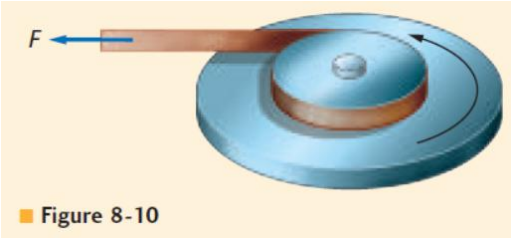


Figure 8-10

24. A 5.8-kg ladder, 1.80 m long, rests on two sawhorses. Sawhorse A is 0.60 m from one end of the ladder, and sawhorse B is 0.15 m from the other end of the ladder. What force does each sawhorse exert on the ladder?

25. What would be the forces exerted by the two sawhorses if the ladder in Example Problem 24 had a mass of 11.4 kg?

26. A 7.3-kg ladder, 1.92 m long, rests on two sawhorses, as shown in Figure 8-15. Sawhorse A, on the left, is located 0.30 m from the end, and sawhorse B, on the right, is located 0.45 m from the other end. Choose the axis of rotation to be the center of mass of the ladder.

a. What are the torques acting on the ladder?

b. Write the equation for rotational equilibrium.

c. Solve the equation for F_A in terms of F_g. d. How would the forces exerted by the two sawhorses change if A were moved very close to, but not directly under, the center of mass?

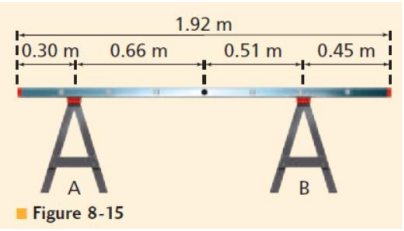


Figure 8-15

27. A 4.5-m-long wooden plank with a 24-kg mass is supported in two places. One support is directly under the center of the board, and the other is at one end. What are the forces exerted by the two supports?

28. A 85-kg diver walks to the end of a diving board. The board, which is 3.5 m long with a mass of 14 kg, is supported at the center of mass of the board and at one end. What are the forces on the two supports?

29. A wheel is rotated so that a point on the edge moves through 1.50 m. The radius of the wheel is 2.50 m, as shown in Figure 8-21. Through what angle (in radians) is the wheel rotated?

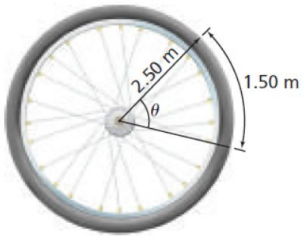


Figure 8-21

30. The outer edge of a truck tire that has a radius of 45 cm has a velocity of 23 m/s. What is the angular velocity of the tire in rad/s?

31. A steering wheel is rotated through 128°, as shown in Figure 8-22. Its radius is 22 cm. How far would a point on the steering wheel's edge move?

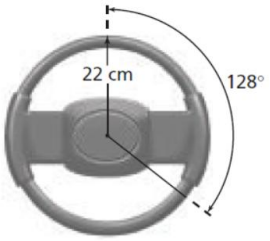


Figure 8-22

32. Propeller A propeller spins at 1880 rev/min.
a. What is its angular velocity in rad/s?
b. What is the angular displacement of the propeller in 2.50 s?

33. The propeller in the previous problem slows from 475 rev/min to 187 rev/min in 4.00 s. What is its angular acceleration?

34. An automobile wheel with a 9.00 cm radius, as shown in Figure 8-23, rotates at 2.50 rad/s. How fast does a point 7.00 cm from the center travel?



Figure 8-23

35. Washing Machine A washing machine's two spin cycles are 328 rev/min and 542 rev/min. The diameter of the drum is 0.43 m.

a. What is the ratio of the centripetal accelerations for the fast and slow spin cycles? Recall that $a_c = v^2/r$ and $v = r\,\omega$

b. What is the ratio of the linear velocity of an object at the surface of the drum for the fast and slow spin cycles?

36. Find the maximum centripetal acceleration in terms of g for the washing machine in problem 35.

37. A laboratory ultracentrifuge is designed to produce a centripetal acceleration of 0.35×10^6 g at a distance of 2.50 cm from the axis. What angular velocity in rev/min is required?

38. Wrench A bolt is to be tightened with a torque of 8.0 Nm. If you have a wrench that is 0.35 m long, what is the least amount of force you must exert?

39. What is the torque on a bolt produced by a 15-N force exerted perpendicular to a wrench that is 25 cm long, as shown in Figure 8-24?

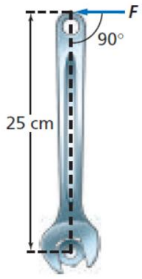


Figure 8-24

40. A toy consisting of two balls, each 0.45 kg, at the ends of a 0.46-m-long, thin, lightweight rod is shown in Figure 8-25. Find the moment of inertia of the toy. The moment of inertia is to be found about the center of the rod.

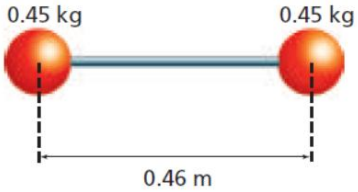


Figure 8-25

41. A bicycle wheel with a radius of 38 cm is given an angular acceleration of 2.67 rad/s^2 by applying a force of 0.35 N on the edge of the wheel. What is the wheel's moment of inertia?

42. Toy Top A toy top consists of a rod with a diameter of 8.0-mm and a disk of mass 0.0125 kg and a diameter of 3.5 cm. The moment of inertia of the rod can be neglected. The top is spun by wrapping a string around the rod and pulling it with a velocity that increases from zero to 3.0 m/s over 0.50 s. a. What is the resulting angular velocity of the top? b. What force was exerted on the string?

43. A 12.5-kg board, 4.00 m long, is being held up on one end by Ahmed. He calls for help, and Judi responds.

a. What is the least force that Judi could exert to lift the board to the horizontal position? What part of the board should she lift to exert this force?

b. What is the greatest force that Judi could exert to lift the board to the horizontal position? What part of the board should she lift to exert this force?

44. Two people are holding up the ends of a 4.25-kg wooden board that is 1.75 m long. A 6.00-kg box sits on the board, 0.50 m from one end, as shown in Figure 8-26. What forces do the two people exert?

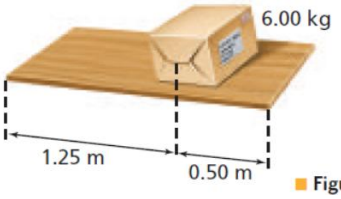


Figure 8-26

45. A car's specifications state that its weight distribution is 53 percent on the front tires and 47 percent on the rear tires. The wheel base is 2.46 m. Where is the car's center of mass?

46. A wooden door of mass, m, and length, *l,* is held horizontally by Dan and Ajit. Dan suddenly drops his end.

a. What is the angular acceleration of the door just after Dan lets go?

b. Is the acceleration constant? Explain.

47. Topsoil Ten bags of topsoil, each weighing 175 N, are placed on a 2.43-m-long sheet of wood. They are stacked 0.50 m from one end of the sheet of wood, as shown in Figure 8-27. Two people lift the sheet of wood, one at each end. Ignoring the weight of the wood, how much force must each person exert?

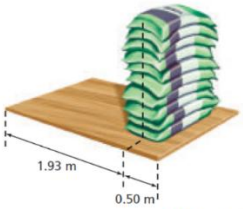


Figure 8-27

48. Basketball A basketball is rolled down the court. A regulation basketball has a diameter of 24.1 cm, a mass of 0.60 kg, and a moment of inertia of 5.8×10^{-3} kgm^2. The basketball's initial velocity is 2.5 m/s.
a. What is its initial angular velocity?
b. The ball rolls a total of 12 m. How many revolutions does it make?
c. What is its total angular displacement?

49. The basketball in the previous problem stops rolling after traveling 12 m.
a. If its acceleration was constant, what was its angular acceleration?
b. What torque was acting on it as it was slowing down?

50. A cylinder with a 50 m diameter, as shown in Figure 8-28, is at rest on a surface. A rope is wrapped around the cylinder and pulled. The cylinder rolls without slipping.
a. After the rope has been pulled a distance of 2.50 m at a constant speed, how far has the center of mass of the cylinder moved?
b. If the rope was pulled a distance of 2.50 m in 1.25 s, how fast was the center of mass of the cylinder moving?
c. What is the angular velocity of the cylinder?

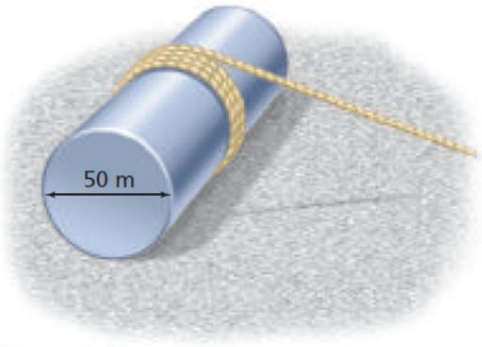


Figure 8-28

51. Hard Drive A hard drive on a modern computer spins at 7200 rpm (revolutions per minute). If the drive is designed to start from rest and reach operating speed in 1.5 s, what is the angular acceleration of the disk?

52. Speedometers Most speedometers in automobiles measure the angular velocity of the transmission and convert it to speed. How will increasing the diameter of the tires affect the reading of the speedometer?

53. A box is dragged across the floor using a rope that is a distance h above the floor. The coefficient of friction is 0.35. The box is 0.50 m high and 0.25 m wide. Find the force that just tips the box.

54. The second hand on a watch is 12 mm long. What is the velocity of its tip?

55. Lumber You buy a 2.44-m-long piece of 10 cm × 10 cm lumber. Your friend buys a piece of the same size and cuts it into two lengths, each 1.22 m long, as shown in Figure 8-29. You each carry your lumber on your shoulders.

a. Which load is easier to lift? Why?

b. Both you and your friend apply a torque with your hands to keep the lumber from rotating. Which load is easier to keep from rotating? Why?

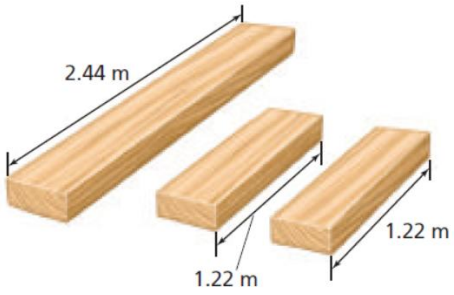
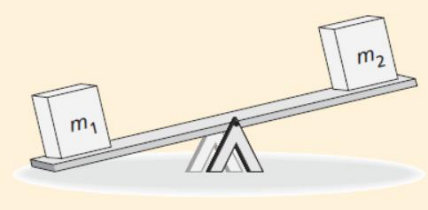


Figure 8-29

56. Surfboard Harris and Paul carry a surfboard that is 2.43 m long and weighs 143 N. Paul lifts one end with a force of 57 N. a. What force must Harris exert? b. What part of the board should Harris lift? 100. A steel beam that is 6.50 m long weighs 325 N. It rests on two supports, 3.00 m apart, with equal amounts of the beam extending from each end. Suki, who weighs 575 N, stands on the beam in the center and then walks toward one end. How close to the end can she come before the beam begins to tip?

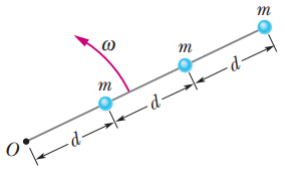
57. The illustration below shows two boxes on opposite ends of a board that is 3.0 m long. The board is supported in the middle by a fulcrum. The box on the left has a mass, m_1, of 25 kg, and the box on the right has a mass, m_2, of 15 kg. How far should the fulcrum be positioned from the left side of the board in order to balance the masses horizontally?



58. In Figure below, three particles of mass m = 0.23kg are fastened to three rod of length d = 0.12 m and negligible mass. The rigid assembly rotates around point O at the angular speed ω = 0.85 rad/s. About O, what are:

a. the moment of inertia of the assembly?

b. the magnitude of the angular momentum of the assembly?



59. A 2-kg uniform sphere with radius of 0.2 m rotates at 6π rev/min. What is the angular momentum of the ball? (Given $\pi^2 = 10$)

60. Big Ben, the Parliament tower clock in London, has hour and minute hands with lengths of 2.7 m and 4.5 m and masses of 60 kg and 100 kg, respectively. Calculate the total angular momentum of these hands about the center point.



(*You may model the hands as long, thin rods rotating about one end. Assume the hour and minute hands are rotating at a constant rate of one revolution per 12 hours and 60 minutes, respectively.*)

61. Complete the following table:

$\theta(rev)$	$\theta(rad)$	$r(m)$	$s(m)$
17	…….	0.5	…….
……..	2	0.25	…….
…….	4	3	12

62. If a torque of 60 Nm is required and the largest force that can be exerted by you is 120 N, what is the length of the lever arm that must be used?

63. A wheel initially turning at 2000 rpm uniformly increases its speed to 6600 rpm in 9.00 seconds. Calculate:

a. the angular acceleration of the wheel in rad/s^2

b. the number revolutions turned by the wheel during the 9.00 seconds interval.

(**Note**: rpm=rev/min=*revolutions per minute*)

64. Calculate the linear speed of the tip of each hand on the following clock:



The length of the second hand is 0.12 m.
The length of the minute hand is 0.15 m.
The length of the hour hand is 0.05 m.

65. Imagine a Ferris wheel that is rotating at the rate of 45 degrees each second.

i. What is the Ferris wheel's period of rotation in seconds?

ii. What is the angular velocity (*angular frequency*) in rad/s and rev/min?



66. A wheel initially turning at 2000 rpm uniformly increases its speed to 6600 rpm in 9.00 sec. Calculate the angular acceleration of the wheel in rad/s^2 and the number revolutions turned by the wheel during the 9.00 sec interval.

67. The blades in a blender rotate at a rate of 7500 rpm. When the motor is turned off during an operation, the blades slow to rest in 3.0 s. What is the angular acceleration as the blades slow down?

68. An automobile engine slows down from 4000 rpm to 1200 rpm in 3.5 s. Calculate (a) its angular acceleration, assumed uniform, and (b) the total number of revolutions the engine makes in this time.

69. If a torque of 55.0 Nm is required and the largest force that can be exerted by you is 135 N, what is the length of the lever arm that must be used?

70. An automobile engine slows down from 4000 rpm to 1200 rpm in 3.5 s. Calculate (a) its angular acceleration, assumed uniform, and (b) the total number of revolutions the engine makes in this time.

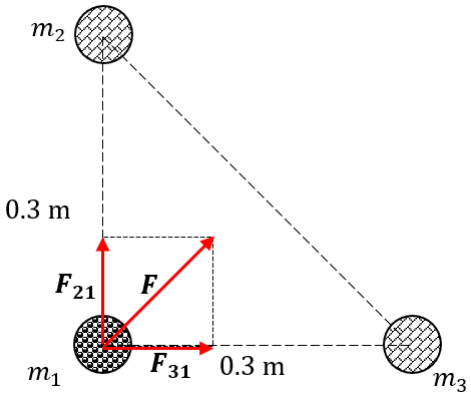
71. The color-enhanced image of Venus shown here was compiled from data taken by *Magellan*, the first planetary spacecraft to be launched from a space shuttle. During the spacecraft's fifth orbit around Venus, *Magellan* traveled at a mean altitude of 361 km. If the orbit had been circular, what would *Magellan*'s period and speed have been?

72. A basketball is being pushed by two players during tip-off. One player exerts an upward force of 15 N at a perpendicular distance of 14 cm from the axis of rotation. The second player applies a downward force of 11 N at a perpendicular distance of 7.0 cm from the axis of rotation. Find the net torque acting on the ball about its center of mass.

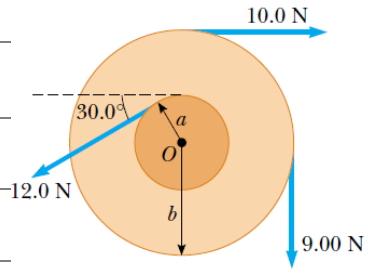
73. Find the moment of inertia of a disc of mass 3 kg and radius 50 cm about the following axes.
a. axis passing through the center and perpendicular to the plane of the disc,
b. axis touching the edge and perpendicular to the plane of the disc and

74. Two masses have a connecting bar of negligible mass. Find the location of the center of mass if (a) m_1 and m_2 are each 5kg, and (b) if m_1 is 5kg and m_2 is 10 kg at (0.2,0.1) and (0.9,0.1) respectively, in meters from the origin.

75. Three 0.30 kg billiard balls are placed on a table at the corners of a right triangle, as shown in Figure below. Calculate the resultant gravitational force F on the ball m_1?



76. Find the net torque on the wheel in Figure below about the axle through O if a = 10 cm and b = 25 cm.



77. Pilots can be tested for the stresses of flying high-speed jets in a whirling "human centrifuge" which take 1.0 min to turn through 20 complete revolutions before reaching its final speed.

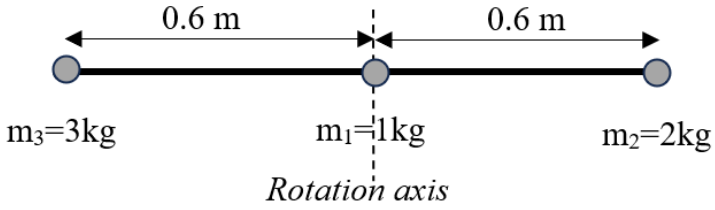
a. What was its angular acceleration (assumed constant)?

b. What was its final speed in rpm?

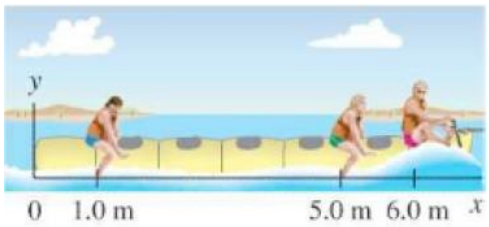
78. In figure below, calculate the moment of inertia I about rotation axis.

-Rod: mass M=1.2 kg, length L=1.2 m

-Point mass: m_1=1kg, m_2=2 kg, m_3=3 kg



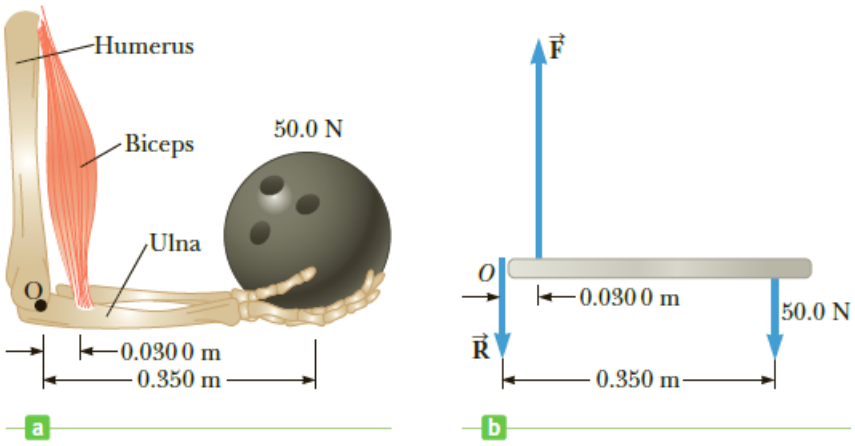
79. Three people of roughly equal masses M on a lightweight (air-filled) banana boat sit along the x axis at positions x_A =1.0 m, x_B=5.0 m, and x_C = 6.0 m, measured from the left-hand end as shown in Figure. Find the position of the center of mass. Ignore mass of boat.



80. A 50 N bowling ball is held in a person's hand with the forearm horizontal, as in Figure (a). The biceps muscle is attached 0.03 m from the joint, and the ball is 0.350 m from the joint.

a. Find the upward force $\vec{F}$ exerted by the biceps on the forearm (the ulna)

b. Find the downward force $\vec{R}$ exerted by the humerus on the forearm, acting at the joint. Neglect the weight of the forearm and slight deviation from the vertical of the biceps.

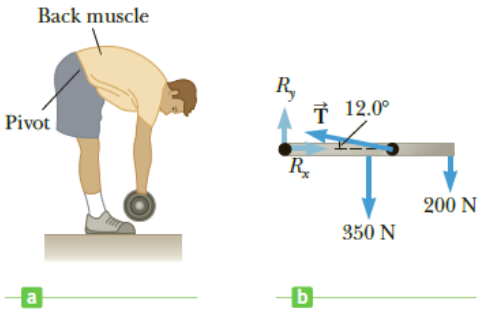


(b) *A weight held with the forearm horizontal* ***(a)*** *The mechanical model for the system*

81. A person bending forward to lift a load "with his back", Figure **(a)**, rather than "with his knees" can be injured by large forces exerted on the muscles and vertebrae. The spine pivots mainly at the fifth lumbar vertebra, with the principal supporting force provided by the erector spinalis muscle in the back. To see the magnitude of the forces involved, and to understand why back problems are common among humans, consider the model shown in **Figure (b)** of a person bending forward to lift a 200 N object. The spine and upper body are represented as a uniform horizontal rod of weight 350 N, pivoted at the base of the spine. The erector spinalis muscle, attached at a point two-third of the way up the spine, maintains the position of the back. The angle between the spine and this muscle is 12.0^0.

a. Find the tension $\vec{T}$ in the back muscle.

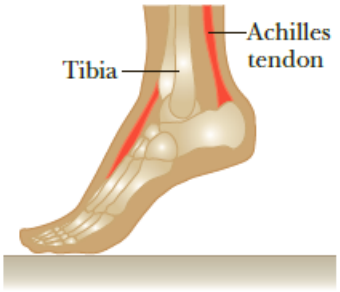
b. Find the compressional force $\vec{R}$ in the spine.



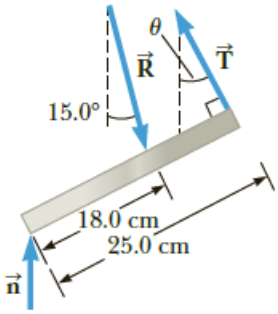
82. When a person stands on tiptoe (a strenuous position), the position of the foot is as shown in **Figure (a)**. The total gravitational force on the body, $\vec{F}_g$, is supported by the force $\vec{n}$ exerted by the floor on the toes of one foot. A mechanical model of the situation is shown in **Figure (b)**, where $\vec{T}$ is the force exerted by the Achilles tendon on the foot and $\vec{R}$ is the force exerted by the tibia on the foot. when $F_g = n = 700$ N,

a. Find the values of T and R

b. Fing the value of angle θ.



a



b

83. The chewing muscle, the masseter, is one of the strongest in the human body. It is attached to the mandible (lower jawbone) as shown in Figure (a). The jawbone is pivoted about a socket just in front of the auditory canal. The forces acting on the jawbone are equivalent to those acting on the curved bar in Figure (b). $\vec{F}_C$ is the force exerted by the food being chewed against the jawbone, $\vec{F}_g$ is the force of tension in the masseter, and $\vec{R}$ is the force exerted by the socket on the man piece of steak with a force of 50.0 N,

a. Find the tension force $\vec{T}$ in the masseter.

b. Find the force $\vec{R}$ exerted by the socket on the mandible.

